

Factory Inventory & Production Management System (MVP)

Project Plan

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Project	Factory Inventory & Production Management System (MVP)
Release	Final academic submission - MVP baseline
Date	24 March 2026
PM Approach	Agile delivery with PRINCE2 Agile-compatible controls over four sprints
Author	Nisal Dulneth Wickramaarachchi (Start-up Manager)
Owner	Uditha Balalla (Project Manager)
Client	Heshan Foods (case organisation)
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Version No	1.0

1 Plan Document History

1.1 Document Location

The controlled copy of this document is stored in the repository at:

“__google drive link__”

1.2 Revision History

Revision date	Previous revision date	Summary of Changes	Changes marked
15 February 2026	N/A	Initial project plan baseline prepared from Sprint 1 kickoff: documented purpose, background, objectives, four-sprint Agile approach, MVP scope, deliverables, team roles, communication cadence, controls, and initial risks.	Yes
06 March 2026	15 February 2026	Refined the plan after Sprint 2 review and manager minutes: confirmed core module delivery, kept MVP scope controlled, and updated the next-phase focus toward deployment readiness, runtime diagnostics, and Vercel-related planning.	Yes
20 March 2026	06 March 2026	Aligned the plan with completed Jira work breakdown and final-sprint coordination artefacts: prioritized release blockers over new scope, clarified submission deliverables, and synchronized documentation/verification work for release closure.	Yes
24 March 2026	20 March 2026	Separated the project plan from the combined Start-up Manager assignment and aligned the content with the implemented repository.	Yes

1.3 Approvals

Electronic acknowledgements are recorded for the submission copy. Formal handwritten signatures are not required for this repository version.

Name	Signature	Title	Date of Issue	Version
Nisal Dulneth Wickramaarachchi	Electronic acknowledgement	Start-up Manager	24 March 2026	1.0
Uditha Balalla	Review acknowledged	Project Manager	24 March 2026	1.0

1.4 Distribution

Name	Title	Date of Issue	Version
Nisal Dulneth Wickramaarachchi	Start-up Manager	24 March 2026	1.0
Uditha Balalla	Project Manager	24 March 2026	1.0
Dulanjali Kulathunga	Scheduling Manager	24 March 2026	1.0
H J M S M Jayasinghe	Quality Manager	24 March 2026	1.0
Heshan Shalinda Koralagamage	Risk Manager	24 March 2026	1.0

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3 Purpose of Document

This document defines the project baseline for the Factory Inventory & Production Management System. It records the business context, objectives, scope, delivery approach, organisation structure, controls, quality expectations, and initial risks so the project can be managed and assessed against an agreed plan.

4 Background

A single food-manufacturing factory requires consistent control over raw-material receiving, production batching, finished-goods availability, and wholesaler dispatching. When these activities are handled through spreadsheets or manual notes, typical issues include stock mismatches, limited traceability for audits, expiry-related waste, and delays in dispatch documentation.

This project introduces a centralised web application for the factory workflow. The implemented solution is a TypeScript monorepo with a React + Vite single-page application, an Express API, Firebase Authentication, MongoDB Atlas persistence, role-based access control, stock movement tracking, FEFO dispatch allocation, delivery-note PDF generation, reporting, audit logging, deployment runbooks, and seed tooling for demo setup.

5 Project Definition

5.1 Project Objectives

- Deliver a working MVP across four timeboxed sprints with a demoable increment at the end of each sprint.
- Provide secure sign-in and server-side RBAC for OWNER, EMPLOYEE, and WHOLESALER roles.
- Support core factory workflows: users, products, suppliers, wholesalers, GRN receiving, BOM definition, production batches, orders, dispatch, reports, and audit review.
- Maintain stock integrity through current-stock snapshots, stock movement records, lot/batch traceability, and controlled state transitions.
- Implement FEFO allocation for dispatch and generate delivery-note PDFs for dispatched orders.
- Provide setup, deployment, and user documentation, plus demo data seeding so the system can be reproduced for review.

5.2 Defined Method of Approach

The project is managed using Agile Scrum across four sprints, with PRINCE2 Agile-compatible controls to keep governance proportionate to the project size. Work is tracked in a backlog prioritised using MoSCoW and reviewed each sprint through planning, daily coordination, a sprint review, and a retrospective.

The Start-up Manager owns backlog readiness, integration sequencing, and demo preparation. Project controls are applied through documented roles, quality gates, risk review, and release verification rather than heavyweight process overhead.

5.3 Project Scope

In scope:

- React SPA client and Express API implemented in TypeScript within a monorepo.
- Firebase Authentication on the client with Firebase Admin token verification on the server.
- MongoDB Atlas persistence for users, products, suppliers, wholesalers, GRNs, BOMs, batches, orders, dispatches, stock snapshots, stock movements, and audit logs.
- RBAC enforcement for OWNER, EMPLOYEE, and WHOLESALER roles on backend endpoints and matching role-aware frontend navigation.
- Operational modules for authentication, users, products, suppliers, wholesalers, GRN, BOM, batches, orders, dispatch, reports, and audit.
- Deployment/runbook documentation, environment templates, and demo-data seeding.

Out of scope:

- Finance and accounting features such as invoicing, payment processing, and ERP integrations.
- Advanced analytics, forecasting, or AI-driven procurement recommendations beyond the implemented reports.
- Credit management workflows and credit-limit automation beyond basic wholesaler master-data support.
- Multi-factory or multi-warehouse operation management.

5.4 Project Deliverables and/or Desired Outcomes

- Frontend single-page application for operational users and wholesalers.
- Backend REST API with validation, RBAC, audit logging, and stock transaction logic.
- Persistence/data layer using MongoDB models for master data and operational records.
- Seed/setup tooling, environment templates, and reproducible local/deployment runbooks.
- Automated tests for FEFO, BOM consumption, integration flows, and basic client behaviour.
- Documentation pack: README, architecture guide, deployment guide, all-in-one setup guide, user guide, sprint artefacts, and role-based assignment documents.

5.5 Exclusions

- Finance and accounting features such as invoicing, payment processing, and ERP integrations.
- Advanced analytics, forecasting, or AI-driven procurement recommendations beyond the implemented reports.
- Credit management workflows and credit-limit automation beyond basic wholesaler master-data support.
- Multi-factory or multi-warehouse operation management.

5.6 Constraints

- The project is timeboxed to four sprints with a small named team.
- The technology stack is fixed to React + Vite, Express API, Firebase Authentication, MongoDB Atlas, and Vercel deployment.
- MongoDB transactions depend on replica-set support in the target Atlas cluster.
- CORS, Firebase service-account formatting, and environment variables must be correct for deployed environments.
- The submission must satisfy academic documentation requirements as well as technical delivery goals.

5.7 Assumptions

- A Firebase project with Email/Password authentication is available for development and demo users.
- A MongoDB Atlas cluster is available and can support the required data volume and transaction model.
- Vercel access is available for frontend and backend deployment, where required for review.
- All team members can access the shared repository and documentation area to update their assigned artefacts.
- Key dependencies remain available: Firebase credentials, MongoDB connection details, npm/Node tooling, and deployment configuration access.

5.8 Legal and ethical

- User, supplier, and wholesaler contact data must be handled responsibly and only exposed according to the assigned role.
- Secrets such as Firebase service-account values and MongoDB connection strings must remain in environment variables and out of version control.
- Audit records, stock movements, batch traceability, and expiry data must be maintained accurately because they affect business accountability and food-handling decisions.
- Third-party libraries and platforms must be used in accordance with their licence terms and acceptable-use conditions.
- Team members are expected to communicate honestly, avoid discriminatory or disrespectful behaviour, and report quality or security concerns promptly.

6 Project Organisation Structure

6.1 Project Management Team Structure

Role	Named owner	Main responsibility
Project Manager	Uditha Balalla	Directed project planning and approvals while contributing to the authentication workstream across access-flow definition, auth handling, backend verification, and onboarding guidance.
Start-up Manager	Nisal Dulneth Wickramaarachchi	Served as lead developer and primary technical integrator, delivering most of the React/Express implementation, including operational modules, stock logic, deployment hardening, and release readiness.
Scheduling Manager	Dulanjali Kulathunga	Managed sprint scheduling and communication while supporting product CRUD delivery across the product-management UI, backend handlers, and related implementation dependencies.
Quality Manager	H J M S M Jayasinghe	Led quality gates and validation reporting while supporting supplier operations through supplier CRUD review, UI checks, validation, and GRN-linked verification.
Risk Manager	Heshan Shalinda Koralagamage	Led risk tracking and mitigation while supporting report-generation work through reporting-flow review, output checks, and deployed-behavior verification.

6.2 Individual Role Descriptions

Project Manager

Uditha Balalla maintained the project baseline, sprint decisions, and approval flow, and supported the authentication workstream by coordinating the Firebase access model, protected-route expectations, and related setup and onboarding guidance.

Start-up Manager

Nisal Dulneth Wickramaarachchi served as the Start-up Manager and lead developer, completing the majority of the system implementation across the React/Express codebase, integrating the main operational modules, and driving deployment hardening, technical alignment, and release readiness.

Scheduling Manager

Dulanjali Kulathunga maintained the sprint calendar, meeting records, and action tracking, while supporting the product CRUD workstream by coordinating delivery checkpoints for the product-management UI, API handling, and related implementation dependencies.

Quality Manager

H J M S M Jayasinghe led the quality plan, acceptance checks, and validation reporting, and contributed supplier-operation support through supplier CRUD review, UI and validation checks, and confirmation that supplier data behaved correctly in the GRN flow.

Risk Manager

Heshan Shalinda Koralagamage maintained the risk register and mitigation tracking, and contributed to the reporting workstream by monitoring report reliability, output consistency, and release-risk impacts during deployment and hardening.

7 Communication Plan

Communication is kept lightweight and traceable. Working decisions are captured through the repository documentation area, sprint artefacts, and meeting records, so the team can review progress without relying on undocumented verbal updates.

Activity	Cadence	Participants	Output	Owner
Daily sync	Working days	All team members	Blockers, progress, and priorities for the next 24 hours	Start-up / Scheduling
Sprint planning	Start of each sprint	All role owners	Sprint goal, prioritised backlog, task owners, and acceptance expectations	Project Manager
Mid-sprint quality checkpoint	Once per sprint	Start-up, Quality, relevant owners	Definition of Done status, defects, and rework actions	Quality Manager
Weekly risk review	Weekly	PM, Start-up, Risk	Updated risk log with owners, mitigations, and escalation decisions	Risk Manager
Sprint review/demo	End of each sprint	Team and reviewer	Working increment demonstration and acceptance observations	Project Manager
Sprint retrospective	End of each sprint	Team	Agreed on process improvements for the next sprint	Scheduling Manager

8 Project Quality Plan

- Definition of Done applies to all sprint deliverables: acceptance criteria met, review completed, validation in place, documentation updated, and demo path prepared.
- Build, test, and lint checks are required before a release candidate is accepted for final review.
- Critical flows are smoke-tested with seeded data: login, GRN confirm, batch completion, order confirmation, dispatch with FEFO allocation and delivery-note PDF, reports, and audit.
- Validation is enforced on the backend using Zod schemas and safe error responses; the frontend uses controlled forms and guided user feedback.
- Documentation changes are treated as part of the deliverable, so setup, deployment, and user guidance remain aligned with the working system.

9 Project Controls

- **Schedule control:** Sprint dates, ceremonies, and action items are maintained by the Scheduling Manager and reviewed with the Project Manager.
- **Progress control:** Backlog items, sprint goals, and delivered increment status are reviewed in daily syncs and sprint reviews.
- **Change control:** When new requests threaten sprint goals, the Project Manager approves re-prioritisation or de-scoping to protect the MVP baseline.
- **Quality control:** Review, validation, build/test/lint checks, and QA sign-off are required before a sprint increment is accepted.
- **Risk control:** Top risks are reviewed weekly with owners and mitigations, with escalation triggered when schedule, deployment, or data integrity is threatened.
- **Release control:** The Start-up Manager coordinates environment readiness, deployment checklist completion, and final smoke verification before demo or submission.
- **Exception process:** If a sprint goal becomes unachievable, the team records the issue, agrees on corrective action, and re-baselines the immediate plan rather than hiding the variance.

10 Initial Risk Log

Ref	Risk	Probability	Severity	Mitigation/response	Owner	Status at initiation
R-01	Firebase credentials or service-account formatting are unavailable for environment setup	Medium	High	Validate credential availability early, keep env examples current, and verify private-key formatting before integration.	Start-up Manager	Open
R-02	Vercel routing or CORS blocks nested /api paths in deployment	Medium	High	Verify serverless routing, exact origin allowlist, and OPTIONS preflight handling; test deployed /api routes early.	Start-up / Risk	Open
R-03	MongoDB Atlas cluster does not provide the transaction support expected by stock mutation flows	Medium	High	Use a replica-set-capable cluster and verify transaction support before relying on production-style flows.	Start-up Manager	Open
R-04	Test coverage is too light to detect release regressions	Medium	Medium	Prioritise FEFO, BOM, integration, and smoke tests for critical workflows each sprint.	Quality Manager	In progress
R-05	Sprint scope exceeds the available time and creates late-stage hardening pressure	Medium	Medium	Use MoSCoW prioritisation and de-scope non-MVP items to protect the essential release path.	Project Manager	Open